Assignment 2 – Part 2: Situated Learning

AIML ZG519 – NLP Applications | Selected topic: **Information Extraction** | Group 86

Student details

Group number	86
Team member names	1. Prajwal Shetty K P (2025AE05434) 2. Neha Yadav (2025AE05463) 3. Poornima Prakash Patil (2025AE05411) 4. Narumalla Pavan Kumar (2025AE05457)
BITS email id	2025ae05434@wilp.bits-pilani.ac.in 2025ae05463@wilp.bits-pilani.ac.in 2025ae05411@wilp.bits-pilani.ac.in 2025ae05457@wilp.bits-pilani.ac.in
Task allocation	**Prajwal Shetty K P:** Part A data prep & EDA; Part D SciBERT fine-tuning; notebook integration **Neha Yadav:** Part B CRF; Part C BiLSTM-CRF **Poornima Prakash Patil:** Part E LLM NER; Part F relation extraction **Narumalla Pavan Kumar:** Part G comparison & error analysis; Part 2 Situated Learning report
Industry / Domain	Scientific publishing and R&D (chemicals / materials / life-science research)
Organisation type	R&D knowledge-management team of a science-based manufacturer or publisher

Status statement (Section 6 of the brief): we distinguish **observed practice** (information extraction that is publicly documented and in production in scientific publishing, cited below) from our **proposed application** for an R&D knowledge-management team. The proposal is evidenced by the working prototype built in Part 1 of this assignment; no confidential organisational data is used – all examples come from the public CrossRE dataset.

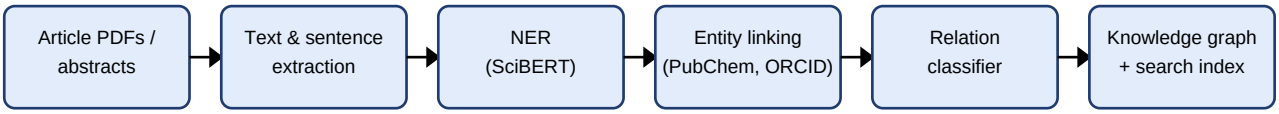
5.1 Industry Context – Anchor

Industry / Organisation	Scientific publishing and corporate R&D. Publishers and public literature services already run large-scale entity extraction (e.g. NCBI **PubTator** annotates genes, chemicals, diseases, species and variants across PubMed; **Europe PMC** exposes text-mined annotations on articles; the open-source **ChemDataExtractor** toolkit extracts chemical entities and properties from papers). Inside a manufacturer, the R&D knowledge-management team curates the internal and licensed article library.
Selected NLP concept	Information Extraction: Named Entity Recognition (NER) + Relation Extraction (RE)
Application / use case	Automatically index every article by the compounds, elements, proteins/enzymes, scientists, theories, awards, institutions and journals it mentions, plus the relations between them (role, origin, part-of, physical, related-to), to power a searchable research index / knowledge graph.
Problem addressed	Researchers cannot quickly answer questions such as *which compounds were studied together with enzyme X* or *which scientists and institutions worked on theory Y*. Keyword search misses synonyms and does not know entity types; manual tagging by librarians is slow, inconsistent and does not scale to thousands of new papers a month.
Users / stakeholders	Bench scientists and project leads (search & discovery); KM team / librarians (curation); IP & patent analysts (prior-art, freedom-to-operate); regulatory & product-safety teams (where a substance is discussed); R&D management (expertise location, collaboration mapping); IT / data-platform team (operations).

5.2 Concept in Practice – Apply

Data / input	Full-text articles and abstracts (PDF/XML) from licensed journals and internal technical reports. Training/evaluation data: an annotated sentence set; our prototype uses CrossRE-science (454 training sentences, 2545 entity mentions, 18 entity types, 17 relation types).
NLP approach / concept	Sequence labelling NER with BIO tags (entity-level strict F1 evaluation); relation classification over candidate entity pairs; LLM prompting as a zero-/few-shot alternative and for pre-annotation.

Tool / model / framework	HuggingFace Transformers (SciBERT `allenai/scibert_scivocab_cased`), sklearn-crfsuite (CRF baseline), PyTorch BiLSTM-CRF with GloVe, Qwen2.5-7B-Instruct via llama.cpp (local LLM), seqeval / scikit-learn for evaluation. Production additions: a PDF-to-text parser (e.g. GROBID), an annotation tool (e.g. Label Studio), a graph store / search index (e.g. Neo4j, OpenSearch).
Workflow / architecture	See diagram: ingest → text & sentence extraction → NER → entity normalisation/linking to identifiers → relation classification for co-occurring entity pairs → triples loaded into a knowledge graph and a faceted search index → researcher UI. Low-confidence outputs are routed to curators, whose corrections become new training data.
Output / result	Per-article JSON records of typed entities and relation triples, e.g. (*scientist*, role, *university*) or (*chemicalcompound*, part-of, *chemicalcompound*), exposed as search facets and graph queries.
Evidence	Prototype from Part 1 (executed notebook, numbers below are copied from its printed outputs) and the saved LLM prompt/response log `llm_ner_outputs.csv`.



Human-in-the-loop: curators review low-confidence entities/relations; corrections are fed back as training data.

Evidence 1 – NER results (strict entity-level, seqeval) on CrossRE-science test data:

Model (Part 1 prototype)	P	R	F1 (100-sent. sample)	F1 (full test)	Inference ms/sent.
CRF	0.6469	0.5753	0.6090	0.5821	1.2
BiLSTM-CRF	0.6679	0.6570	0.6624	0.6507	2.4
Transformer (SciBERT)	0.7126	0.7514	0.7314	0.7049	40.3
LLM zero-shot	0.7613	0.5499	0.6386	n/a *	8483.9
LLM few-shot	0.7280	0.6606	0.6927	n/a *	11019.6

* The LLM was evaluated only on the shared 100-sentence test sample (about 8–11 s per sentence on CPU makes the full 400-sentence test impractical), so it has no full-test F1. Compare all five models on the 100-sentence sample column.

Evidence 2 – relation extraction (gold entities, 7 classes): logistic-regression macro-F1 = 0.459; fine-tuned SciBERT with typed entity markers macro-F1 = 0.523.

Evidence 3 – representative input/output (few-shot LLM, raw response saved in the CSV):

```
Input: Viral TK phosphorylates aciclovir into its monophosphate form , which is subsequently phosphorylated to active aciclovir triphoshate by cellular kinase s , thus selectively inhibiting viral DNA polymerase .
```

```
Output: [{"text": "Viral TK", "type": "enzyme"}, {"text": "aciclovir", "type": "chemicalcompound"}, {"text": "aciclovir triphoshate", "type": "chemicalcompound"}, {"text": "cellular kinase", "type": "enzyme"}, {"text": "viral DNA polymerase", "type": "protein"}]
```

5.3 Industry Analysis – Analyse

Key benefits / value	(1) **Findability:** type-aware facets (compound, enzyme, scientist, award) replace keyword guessing. (2) **Scale & consistency:** a fine-tuned model tags every new article in milliseconds (measured 40 ms/sentence on CPU for the best model) with the same guidelines every time. (3) **Discovery:** relation triples reveal co-occurrence and provenance networks (who worked where, which compound is part of which material). (4) **Re-use:** the same index feeds patent analytics, regulatory look-ups and RAG assistants.
Limitations / problems	(1) **Accuracy ceiling with little data:** with only 454 annotated sentences the best model reached F1 = 0.731 on the shared sample, so outputs need human review before being trusted for IP or safety decisions. (2) **Long-tail types** (theory, book, chemical element) have very few examples and are the most error-prone. (3) **Ambiguity:** scientist vs. person, organisation vs. university, misc catch-all. (4) **Relations are harder than entities** (best macro-F1 0.523). (5) Prompted LLMs are cheap to start but slow on CPU, sometimes return unparseable JSON and can invent entity types. (6) PDF parsing errors (tables, formulae, ligatures) propagate downstream.

Evidence of effectiveness / impact	In the prototype (F1 on the shared sample: CRF 0.609, BiLSTM-CRF 0.662, Transformer (SciBERT) 0.731, LLM zero-shot 0.639, LLM few-shot 0.693) the fine-tuned SciBERT was the most accurate model at 40 ms/sentence, while the few-shot LLM came second without any training data, beating the CRF and BiLSTM-CRF but at 11.0 s/sentence. This supports a fine-tuned model in production, with the LLM used for bootstrapping annotations. Publicly, services such as PubTator and Europe PMC demonstrate that automated entity annotation is viable at literature scale and is used by curators and researchers for search and database curation.

5.4 Reflection and Future Scope – Reflect

Major challenge / gap	**Annotation scarcity for the organisation's own entity types and relations.** Generic scientific NER covers common types, but each company cares about its own materials, product codes and internal projects; there is little labelled data, and rare types/relations score lowest. Entities are also not yet linked to canonical identifiers, so synonyms (e.g. 'ATP' vs 'Adenosine triphosphate') remain separate nodes.
Recommended improvement	An **active-learning annotation loop**: use the local LLM (few-shot) to pre-annotate new articles, have curators correct them in an annotation tool, and periodically re-fine-tune SciBERT on the growing set; add an **entity-linking** step to PubChem/ChEBI/ORCID/ROR identifiers. This converts cheap LLM suggestions into high-quality training data while keeping the fast, accurate fine-tuned model in production.
Potential future use	A knowledge-graph-grounded **research assistant (Graph-RAG)**: researchers ask natural-language questions ('which catalysts containing ruthenium were reported with yields above 90 %?') and get answers with citations to the extracted triples and source sentences; plus alerting when new papers mention a watched compound, competitor or scientist.

Acknowledgements and sources

Dataset: CrossRE (Bassignana & Plank, 2022; HF `DFKI-SLT/cross_re`). Models: SciBERT (Beltagy et al., 2019), GloVe (Pennington et al., 2014), Qwen2.5-7B-Instruct (Qwen team, 2024). Public systems referenced: NCBI PubTator (Wei et al.; PubTator3, 2024), Europe PMC text-mined annotations, ChemDataExtractor (Swain & Cole, 2016). Tools: HuggingFace Transformers, sklearn-crfsuite, llama.cpp, reportlab. AI assistance: Claude (Anthropic) was used to help draft code and text; all numbers are taken from our executed notebook.